

ASSIGNMENT BRIEF

HTU Course No: 40201430	HTU Course Name: Database Programming
BTEC Unit Code:	BTEC UNIT Name:



Student Name/ID Number/Section	
HTU Course Number and Title	40201430 Database Programming
BTEC Unit Code and Title	
Academic Year	2025-2026 1
Assignment Author	Samir Tartir
Course Tutor	Samir Tartir
Assignment Title	Vehicle Management System
Assignment Ref No	1
Issue Date	10/11/2025
Formative Assessment dates	From 10/11/2025 to 15/01/2026
Submission Date	24/01/2026
IV Name & Date	Samer Suleiman 09/11/2025
Submission Format	
The submission is in the form of a soft copy to be submitted through the university's eLearning system within the deadline specified above from the below link: https://elearning.htu.edu.jo Submission for this assignment is expected to be as follows: 1. An individual written report that is uncompressed and without page borders follows the below guidelines, and contains the sections described in the assignment brief (.docx). 2. The source code of the project (one .c file). 3. Student declaration form filled out and signed properly (.docx). Report guidelines: In a report, you should make use of headings, paragraphs, and subsections as appropriate. In your report, you should make use of headings, paragraphs, and subsections as appropriate. The expected word limit is 5000-10000 words, although you will not be penalized for exceeding the total word limit, do your best to be within the word limit. Your report should be: 1. In a word form of a soft copy submitted to the instructor. 2. Written in a formal business style using single spacing and font size 12. 3. Must be supported with research and referenced using the Harvard referencing system.	
Unit Learning Outcomes	
LO1 Get a high-level understanding of the major DBMS components and their functions. LO2 Develop the required functionalities at the main DBMS components. LO3 Explain and use the principles of concurrency control and transaction management. LO4 Use modern methods to wrap developed DBMS functionalities in usable interfaces and apply the required testing.	
Assignment Brief and Guidance	

You've recently joined the backend systems team at HTU, located inside King Hussein Business Park (KHBP). The university and the business park together manage hundreds of vehicles daily, including those belonging to students, faculty, staff, tenants, and visitors. To improve efficiency and security, HTU is launching a new initiative to develop a custom Vehicle Management System (VMS).

Your team is responsible for designing and implementing the database management system (DBMS) that will support this VMS by storing data in CSV files. Depending on the subcomponent you choose to implement, the system must manage the different operations of a VMS, such as vehicle registrations, access permissions, parking reservations, visitor approvals, and movement histories. Additionally, the DBMS must handle concurrent requests such as multiple vehicles attempting entry, overlapping parking bookings, or simultaneous visitor check-ins, ensuring that the system remains consistent and accurate as many users interact with it at the same time.

You must design and implement the DBMS that will store, retrieve, and manage all vehicle-related data, as well as the concurrency mechanisms required to handle simultaneous operations. The system will be used by all types of users to perform the different actions they need, and by the platform itself to track logs.

Choose **one** of the following subcomponents to implement:

- Gate Access and Vehicle Authorization System – controls entry permissions and gate validation.
- Smart Parking Allocation and Reservation System – manages slot availability, reservations, and usage history.
- Visitor Vehicle Management System – handles visitor approvals, temporary access credentials, and incident logging.

Part 1:

1. Provide a well-defined problem statement with a clear set of user and system requirements. Then design a relational database system by providing the physical model of the database using appropriate design tools. The design should contain at least four interrelated tables. **(Report)**
2. Provide a full description the architecture of the DBMS and the function of the sub-components along with their programming techniques. The targeted programming techniques that were used in this component such as different types of data structures. You need to mention which components you decided to use in your DBMS implementation. **(Report)**
3. Study the ability to transfer data between different DBMS components and evaluate it using appropriate diagrams, data flow diagrams to show data movement through the system components and flowcharts to describe how the system works. **(Report)**
4. Study and report the effectiveness of the database and DBMS design in terms of user and system requirements. **(Report)**

Part 2:

1. Implement the CRUD statements on each table in C. Note: the CRUD statements should act like the original SQL statement. For example, the insert statement will not add the new record if the primary key is duplicated, or if the foreign key is invalid. **(Report and Project)**
2. Implement the storage and buffer managers using the appropriate data structures. The reserved memory needs to be dynamically allocated during the in/out communication with the data structure through the storage manager. **(Report and Project)**
3. Optimize the performance of the system using an indexing technique . **(Report and Project)**
4. Provide a critique of the process of mapping SQL queries (in standard DBMS) to the developed CRUD statements (in your system) and suggest improvements to optimize the developed CRUD queries. **(Report)**

Part 3:

1. List and provide a detailed description of the top standard DBMS concurrency control techniques. **(Report)**
2. Pick one of the techniques described, and implement it in your system to handle concurrent database operations. **(Report and Project)**

Part 4:

1. To allow using the system, provide an interface that exposes all offered DBMS functionalities. The interface can be advanced (GUI) or basic (CLI). **(Report and Project)**
2. Provide a testing plan and show its results in ensuring the compatibility and functionality of developed DBMS components and that they satisfy the requirements. **(Report and Project)**
3. Discuss your DBMS implementation in terms of areas of improvements needed to ensure the continued effectiveness of any database **(Report)**

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Get a high-level understanding of the major DBMS components and their functions.	P1 Design a relational database system using appropriate design tools and techniques, containing at least four interrelated tables, with well-defined problem statements and a clear set of user and system requirements.	M1 Evaluate the ability to transfer data between different DBMS components including diagrams to show data movement through the system components and flowcharts describing how the system works.	D1 Evaluate the effectiveness of the design in terms of user and system requirements.
	P2 Describe in detail the architecture of the DBMS and the function of the sub-components along with their programming techniques.		
LO2 Develop the required functionalities at the main DBMS components.	P3 Write the SQL queries based on the user requirements and implement the corresponding CRUD statements in the language of development.	M2 Implement any type of indexing technique to optimize the performance of the simulated SQL queries.	D2 Criticize the process of mapping SQL queries to the developed CRUD statements and suggest improvements to optimize the developed CRUD queries.
	P4 Develop the required dynamic data structure at the storage and buffer managers.		
LO3 Explain and use the principles of concurrency control and transaction management.	P5 Research the standard DBMS concurrency control techniques and their usage in the transaction manager.	M3 Implement the parallel data processing functionality to handle concurrent database operations.	D3 Evaluate the DBMS in terms of Improvements needed to ensure the continued effectiveness of any database.
LO4 Use modern methods to wrap developed DBMS functionalities in usable interfaces and apply the required testing.	P6 Develop the needed interface that is able to handle all offered DBMS functionalities.	M4 Suggest and apply a testing plan to ensure compatibility and functionality of developed DBMS components.	